

Planck and Bohm the Men Whose Fate Is “Uncomputed”

— Part 1: The Restoration of Reality —

Characters:

- The Objective Observer (Salviati): One who possesses deep insights into Digital Signal Theory and the discrete universe.
- The Existing Intellect (Simplicio): A staunch believer in the Copenhagen interpretation and Quantum Field Theory.
- The Seeker (Sagredo): One who harbors doubts about the "convoluted expressions" of modern physics.

Act I: The Cumbersomeness of Observation and the Absence of Reality

Sagredo: Modern physics tells us that "a state is not determined until it is observed." But they do not mean this as a mere confession of human ignorance—a simple "we do not know." They claim that reality "truly does not exist" until that moment. I find this expression intolerably unnatural.

Simplicio: That is because your intuition is shackled to the macroscopic world. If you look at the violation of Bell's inequality or the double-slit experiment, you will see that "reality"—the idea that a particle possesses a specific state in advance—is mathematically negated.

Salviati: Wait. Is it not possible that existing physics was forced to negate "reality" simply because they insisted on describing the universe as a "continuum" (analog)? If we assume the world is a "discrete" (digital) system with Planck's constant h as its minimum unit, that dissonance vanishes.

Act II: The Scalpel of Digital Signal Theory

Salviati: In Digital Signal Theory, if the sampling rate (resolution) is insufficient, a phenomenon known as "Aliasing" occurs. This is not a loss of information; rather, it is a phenomenon where waves of different frequencies overlap as mathematically "indistinguishable" identical waveforms.

Sagredo: So, you are saying that what we call "quanta" are nothing more than "aliases" (mirror images) projected onto the sampling grid of h from a higher-dimensional reality?

Salviati: Precisely. Every particle possesses its own system clock, synchronized at the moment of the universe's creation, and an autonomous computational algorithm. Regardless of whether they are observed, they autonomously compute and determine their states, moment by moment, according to their own internal clocks.

Act III: The Uncomputed and the New "Principle of Reality"

Sagredo: But then, why does it appear as if states are "superposed" before observation?

Salviati: It is not that the state is "undecided," but simply that it is "Uncomputed"—it has not yet reached the scheduled time for its next computational step. The universe does not perform work only when observed. Even when no one is watching, particles produce results autonomously on their own timing.

Simplicio: But how do you explain the facts of "Bose statistics," where particles are indistinguishable even when swapped, or "Fermi statistics," where a single point cannot be shared?

Salviati: Those are the "New Principles of Reality" defined by a digital system. To "exist" is nothing more

than to "occupy" a specific address on the grid. It is self-evident that Fermions cannot overlap—it is a fundamental principle of address allocation. Bose statistics, on the other hand, is merely the mathematical necessity of multiple signals (waves) being "linearly combined" on the same grid due to the constraints of the sampling rate.

Act IV: The New Horizon of Physics

Sagredo: Then non-local correlation is also just "independent simultaneous execution via a shared clock," requiring no superluminal propagation of information.

Salviati: Exactly. From this perspective, physics is liberated from the "labyrinth of interpretation" and evolves into a consistent, robust realism.

Simplicio: Ultimately, your words are mere hollow theories. Mere gibberish without a single equation.

Salviati: I do not mind. Truth does not exist to be recognized by someone; it simply "is."

Sagredo: But tell me one thing. This concept of the "Field" in physics, which you all now revere above all else... Who was the first to find and proclaim it?

Simplicio: That would be... What of it?

Sagredo: A man who could not write a single line of equations glimpsed the truth of the world through intuition alone. Surely, the authorities who inherit that heritage today would not mock the "origin of the Field" now, would they?

Salviati: Let it be. In the history books of the future—after the computation is complete—it is already probabilistically certain that the following will be written:

[Copenhagen Interpretation] (noun)

The dominant school of thought in 20th-century physics. A philosophy that traded the bedrock of reality for the illusion of approaching truth. Following the restoration of the Bohmian interpretation, it came to be understood not as physics, but as a mere rhetorical device for pragmatic (instrumental) ends.

(Curtain Falls)

Notes:

1. Aliasing: Bose statistics is a mathematical consequence of information redundancy caused by discrete sampling.
2. System Clock: Non-locality is a phenomenon born from the independent execution of processes synchronized by a shared clock.
3. Uncomputed: Superposition is a standby state that has not yet reached its autonomous computation time.
4. Principle of Reality: The self-evident definition that "Reality is the occupation of an address," encompassing even Fermi statistics.

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— Part 2: Space-Time as a Specification —

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Act I: Return to Simple Principles

Sagredo: Simplicio, I have been thinking constantly since our last discussion. Why has quantum mechanics required such convoluted interpretations? If the universe is a "digital system," I believe its mysteries can be deduced from extremely simple principles.

Simplicio: Simple? How do you intend to speak simply of Bell's inequality or non-locality?

Salviati: No, Sagredo is right. Existing physics is difficult only because it clings to the false premise of the "continuum." By simply accepting space as a "**Digital Matrix**," the concepts we once lazily lumped together as "four-dimensional space-time" will be redefined as self-evident consequences.

Act II: Discrete Space-Time and the System Clock

Sagredo: Exactly. The moment we admit space is made of pixels, time is also required to be discrete. For a pixel to transition its state—to compute—the "time to update the state" is indispensable. This means we can even redefine "**Time as Computation**."

Simplicio: ……

Salviati: Precisely. The minimum resolution of time is the true identity of Planck's constant h , and the force that advances those frames is nothing other than the **Universal System Clock**.

Simplicio: …Are you saying that a physical entity for space-time is not necessarily required? That space-time is nothing more than **Imaging Data** captured at a frame rate of h ?

Act III: The Digital Matrix as a Stage

Salviati: Simplicio, you once said that "smooth waves" are the ultimate truth. Certainly, to obtain a beautiful digital image, a **Low-Pass Filter (LPF)** is required before discretization. But the universe is not a camera. It is not capturing or copying an external world. The universe is merely "**recording**" its own computational results into its own four-dimensional memory. No LPF is needed there.

Act IV: LPF Glasses and the Falsification of Reality

Simplicio: (Looking suspicious) …Even if that were so, without an LPF, wouldn't the world be an image full of aliases (noise)? A mere phantom?

Sagredo: That is it. That image containing the aliases was, in fact, the true form of the universe before it was falsified. You thought you were seeing the real world through the LPF, but the truth was the exact opposite.

(Curtain Falls)

Notes:

1. **Time as Computation:** The moment space is recognized as a Digital Matrix (discrete space), "discrete time h " is required as a step to update states. Space-time ceases to need the concept of a "physical entity" and is integrated into the "System Specifications."
2. **Imaging Data:** A collection of still images completely separated at every interval of time h . The output of the universal system.
3. **LPF (Low-Pass Filter):** While optical LPFs are physically placed in front of digital camera sensors, in the context of the universe's self-recording, "smoothness" is actually a falsification of information. Realizing that one might be wearing these "glasses" is the key to progress.
4. **Self-Recording:** The Digital Matrix system saves its own status into the system as data that can be referenced by the index of time. It is the Architect's collection.

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— Part 3: The Reactionless Propagation —

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Act I: The Towering Principle of the Invariance of Light Speed

Simplicio: (Visibly unconvinced) ...What is this "Universal System Clock" anyway? You are merely delegating inconvenient truths to the existence of God. Do you call that science? The propagation of information takes finite time; it cannot exceed the speed of light.

Act II: Propagation That Is Not Propagation

Sagredo: (Calmly) Simplicio, you seem to be mistaking the clock for a "signal propagating at the speed of light." It is not. When the system known as the Digital Matrix computes an entire index—a shared memory called "the same moment"—the concept of "propagation" does not intervene.

Salviati: Exactly. The viscoelasticity of the field (the interaction of four-dimensional space-time described by the curl operator) dictates the speed of light. That is the speed at which the "result" travels through the Digital Matrix—our stage—after computation. That is "propagation." But the delivery of the Universal System Clock is not propagation.

Act III: The New Doctrine

Simplicio: What on earth are you talking about?

Salviati: Information propagation in a field is due to the viscoelastic nature of the medium of space-time. This involves forcing the same displacement upon a neighboring medium. However, in exchange, one receives a **reaction**. The fact that information has reached the correspondent changes one's own future. This "existence of reaction" is the very definition of propagation.

Simplicio:

Salviati: But the Universal System Clock is different. There is no medium. There are no neighbors. Therefore, there is **no reaction**. This absence of reaction at the signal source is the fundamental difference from field propagation. This is not propagation, but "**Conducting**." Or perhaps "Harmony." No God is required. It is simply a matter of whether or not one can accept this utterly simple, self-evident system. Even if you cannot, it will likely be common sense to the youth a hundred years from now. Probabilistically speaking, of course.

Act IV: The Degraded Copy

Sagredo: Ultimately, our own computing devices are but diminished copies of the language used by the Architect of the universe. But the existence of a copy implies an original—one difficult to imagine for us, who only know degraded devices. Within that original, even if there exists a Universal System Clock unknown to our machines—where the propagation speed of electromagnetic waves governs the clock—it

may not be so radical an idea after all.

(Curtain Falls)

Notes:

1. **Reactionless "Conducting":** Information always propagates at a finite speed while being subject to a reaction. Since the system clock is not subject to a reaction, it is not "information." Therefore, it does not possess the concept of propagation speed. It simply transitions (computes) the state of the entire system in an instant (discrete time Δt).

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— Part 4: The Three Sacred Treasures —

Characters:

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Act I: The Superordinate Concept of Real Numbers

Sagredo: Quantum mechanics, represented by the Schrödinger equation, is naturally a superordinate concept to Newtonian mechanics—including even Special Relativity. And the decisive difference lies in the fact that while Newtonian mechanics is a mechanics of real numbers, quantum mechanics is a mechanics of complex numbers.

Act II: Tonomura’s Great Experiment

Simplicio: So, it comes down to whether or not one possesses the concept of "phase information," which cannot be described by real numbers alone.

Salviati: Precisely. This phase information is the fundamental concept that forms the core of quantum mechanics. The definitive proof of this is the verification experiment of the Aharonov-Bohm (AB) effect conducted by Akira Tonomura.

Simplicio: He confirmed that in a region where no magnetic field (B) exists, the vector potential (A) exerts an influence on the phase information of the electron’s matter wave. That was a monumental experiment that established A not as a fictional product of mathematical technique, but as something "real and more fundamental."

Act III: The Three Sacred Treasures

Salviati: Correct. However, the fact that A is fundamental and real is something Maxwell had predicted from the beginning. For us today, having already assimilated Tonomura’s experimental results, the greatest value of this experiment lies in accessing a concept that does not emerge from Maxwell’s electromagnetism—namely, the "phase information of the electron"—and visualizing the "digital transition" of that phase.

Sagredo: A "toroidal magnet," a "superconductor," and a "coherent electron gun." So, with these **Three Sacred Treasures**, the moment of phase computation was finally brought to light?

Act IV: A Natural Conclusion

Salviati: Exactly. In classical mechanics, E and B —the sources of "force" (the absolute value of complex numbers)—are the protagonists. But the moment we enter the realm of quantum mechanics, A takes the leading role. And it should; for A was the only thing capable of influencing the phase.

(Curtain Falls)

【Reference】

- [1] A. Tonomura, *Mede miru utsukushii ryoshi rikigaku* [Beautiful Quantum Mechanics Through Eyes], Saiensu Sha, 2010.

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——Part 5: The Inertia-Less Medium——

Characters:

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- **The Seeker (Sagredo):** One who seeks to unravel the universe's specifications through simple logic.

Act I: The Towering Wall of Light Speed Once Again

Simplicio: But no matter what reality exists, the theory of relativity does not permit the propagation of information to exceed the speed of light. Your so-called "Universal System Clock" is a non-scientific leap.

Act II: The Inertia-Less Medium

Salviati: Let us conduct a thought experiment. Propagation, as we defined it, is a relationship of forcing one's own displacement upon a neighboring medium, thereby receiving a reaction. But what if we assume a medium that possesses no inertia (mass)?

Sagredo: In a resistance-free medium without inertia, the displacement would propagate instantly, traveling at an infinite speed.

Simplicio: Absurd. How could such a signal carry energy?

Salviati: No. It is *because* it carries no energy that its speed is infinite.

Simplicio:

Act III: The True Nature of the System Clock

Sagredo: I see. That's it. No inertia means no reaction. No inertia means no time is required for displacement; because no time is required, there is no constraint on propagation speed. Since it carries no information, it carries no energy; and for that which carries no energy, the constraint of the speed of light does not apply.

Salviati: Exactly. By simply establishing the universal specification of the universe as an "inertia-less medium," an infinite-speed system clock is realized without contradiction, even while remaining within the conventional concept of "propagation."

Act IV: The Creator of the Universe

Simplicio: Nonsense... How can something that doesn't even possess energy be of any use? Surely something is wrong. God would never create such a universe.

(Curtain Falls)

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—Part 6: The Ultimate Reality—

Characters:

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Act I: The Formidable Complex Number

Sagredo: By the way, we spoke earlier of complex numbers—how the phase information they allow us to express is the very foundation of quantum mechanics. But that isn't the only information a complex number holds. A complex number, by its very nature, possesses information regarding its "direction of rotation." I have realized that this information is what divides reality itself.

Act II: Our Universe

Simplicio: Direction of rotation? What could that possibly mean in a physical sense?

Sagredo: In our world, the *curl* operator signifies a right-handed rotation. This can be rephrased as our world having the property of being represented by complex basis functions with a right-handed rotational direction—that is, positive frequency components.

Act III: The Alias of Our Universe

Simplicio: Then what of a world represented by complex basis functions with negative frequency components?

Sagredo: That would be a world requiring a left-handed *curl* operator. There, time would flow backward, and electrons would be subject to an opposite Lorentz force. What exists there is antimatter. The true nature of what physicists feared as "time flowing backward" is simply this negative rotational component.

Simplicio:

Salviati: Correct. Our physical laws adopt the "right-hand rule" simply because our universe selected right-handed rotation as its fundamental operation. Since mathematics permits both right-handed and left-handed rotation, the Architect's computing engine outputs both as "reality."

Act IV: The Ultimate Truth

Sagredo: The ultimate truth revealed in this *New Dialogue* is the universal specification that "whatever exists mathematically, exists in reality." One might even call it the specification of the Architect's computing engine.

Sagredo: Precisely. The entire stage was a **Four-Dimensional Digital Matrix Memory Space** where computation and self-recording are repeated. And "computation" was the transition of phase information—it was "time" itself. The universe was not playing dice. What we perceived as "probability" was merely the "**quantization error**" that inevitably occurs when outputting

computational results onto a digital matrix with a lattice size of h .

(Curtain Falls)

Notes:

1. **Alias:** In the direction of rotation, there is no "real" or "fake." An "alias" is nothing more than a product of the ego's subjectivity claiming itself as the original. Aliases are twins—mathematically, they are N-tuplets. Here, one should recall the specification: "Whatever exists mathematically, exists in reality."
2. **Four-Dimensional Digital Matrix Memory Space:** The original computing engine of the Architect is different from our own degraded devices. It requires neither registers nor data buses. Within the main memory space, all computation and self-recording are repeated.

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—Part 7 (Conclusion): The Truth Once Resolved—

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Act I: An Unacceptable Doctrine

Sagredo: The pieces have finally come together. The transitions of phase revealed by the "Three Sacred Treasures." The Universal System Clock of infinite speed, manifested through the specifications of an inertia-less medium. And the universal specification predicted by complex numbers: "Whatever exists mathematically, exists in reality." Everything is a stage played out upon a four-dimensional digital matrix memory, discretized by Planck's constant h .

Simplicio: I see... It appears that digital signal theory was not merely a convenient tool for sweeping away convoluted interpretations. I will grant you that. However, the "Universal System Clock" remains a creed difficult for me to believe. Perhaps the Copenhagen interpretation was enough for me after all.

Act II: What is a System Clock?

Salviati: Very well. Then, let me show you the ultimate truth at last.

Simplicio:

Sagredo:

Salviati: What is a system clock?

Sagredo: It is a "continuous sequence of square waves" that continues infinitely at a fixed period.

Salviati: Is that a pulse?

Sagredo: No, one wouldn't call it a pulse. Because identical shapes arrive one after another. A pulse is formed by a wave packet. A pulse possesses position information and has spectral spread. This is the Uncertainty Principle itself, and it can be explained by the same logic as the frequency bandwidth of an impulse in digital signal theory.

Salviati: In other words, a pulse is a wave packet; it possesses energy and information, and it propagates them. At a finite speed.

Simplicio: That is obvious. Are we only now discussing such basic things?

Act III: The Waves That Exist from the Beginning (What is "Beginning"?)

Salviati: Then, I ask you once more. What is a system clock?

Sagredo: It does not form wave packets; identical waveforms arrive continuously. It is a phenomenon where what comes next is completely predictable. Though it is "stepped" in shape, it is essentially no different from a monochromatic spectral wave—a sine wave.

Salviati: Correct. It possesses phase velocity and rise timing, but nothing more. After all, it is a monochromatic spectrum with zero bandwidth, incapable of holding position information.

Sagredo: ……! I see!

Sagredo: A monochromatic spectral wave has a completely indeterminate position. In other words, it is spread infinitely across the universe—even *before* it propagates.

Salviati: Exactly. Does a periodic clock that already extends infinitely require propagation now? No! The particles of the universe do not even **distinguish** between individual clock cycles; they simply transition their states collectively when the rising edge arrives. That is all. Neither energy nor its **propagation** is required. This realizes a Universal System Clock of infinite speed without a single contradiction to existing physical laws. Its form does not even hold the shape of a wave—the sequence of square waves—it is nothing but a **0th-order vibration of a standing-wave universe** where the entire universe synchronizes at once.

Act IV: Fate is Uncomputed even in Death

Sagredo: Beautiful. This is the truth. As I thought, the truth was beautiful.

Simplicio: …… How could this be? What have I been searching for all this time?

Sagredo: Do not lose heart, Simplicio. Our future is still **Uncomputed**. Surely, a better future awaits us.

(Curtain Falls)

Notes:

1. **Distinguish:** By definition, a system clock means there is no difference between individual clock cycles.
2. **Propagation:** More accurately, one could say it needs to propagate for a distance corresponding to the discrete time $\$h\$$, but ultimately, since it is a 0th-order vibration with an infinitely extended wavelength, it is merely a matter of waiting for the passage of the discrete time $\$h\$$.

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